Autonomous Robot Sorting

Team: _____ Names: _____ 22 October 2026

Mission: Observe, decide, act, check. Sort red and blue objects. Reject unknown or missing readings. Work in pairs: one types, one predicts; swap roles after each activity.

A. First Python experiments (25 min)

Run `python exercises/01_python.py`. Predict first: how many greeting lines? ____

Change the greeting and make it repeat five times. Actual greeting lines: ____

Change the color decision so red goes right and blue goes left. Explain your `if`: _____

B. Sensors and decisions (35 min)

Run `python exercises/03_vision.py`. Predict what red / blue / unknown produce.

Try an unfamiliar color and a missing reading. What should happen? _____

Compare the image examples or live camera. One recognition problem: _____

C. Robot, conveyor and camera (3 × 15 min)

Follow your assigned rotation. Keep hands clear of moving equipment. The instructor enables physical motion. Offline: `python exercises/02_robot.py` and `python -m workshop`.

One observation: _____ One improvement: _____

D. Build your sorter (45 min)

Run `python exercises/04_sort.py`. Use the five-object sequence shown in the file.

Predict: left ____ right ____ reject ____ Actual: left ____ right ____ reject ____

Reverse red and blue destinations; run again. Does unknown stay rejected? ____

Run `python exercises/05_visual_sort.py`; open `sorting-result.html` in a browser.

Change a Python rule and run again. Which block moved differently? _____

E. Challenge board (40 min)

Finish 1 and 2 first, then choose. Tick a box after you can explain your result.

[ ] 1 Easy: Reverse the destinations and verify the result.

[ ] 2 Easy: Count left, right and reject. Do they sum to the number of objects?

[ ] 3 Medium: Add green and choose a destination. Test it.

[ ] 4 Medium: Reject missing (`None`) and invalid color readings.

[ ] 5 Medium: Create two sequences. Predict counts before running each one.

[ ] 6 Advanced: Improve reliability. Demonstrate your idea in simulation.

Show your result (2–3 min per pair)

Our rule: _____ Test result: _____

Sensor: _____ Actuator: _____ Failure and response: _____

Stop and ask the instructor if equipment behaves unexpectedly. No installation or ROS commands are needed for these activities.